Report

CSA2001: Fundamentals in AI and ML

Submitted by: Name: Shiwendra Shiwam Jha

Registration Number: 25BAI10357

Branch: B.Tech CSE AI and ML

Course Code: CSA2001 – Fundamentals in AI and ML

I. Introduction & Problem Statement

In modern healthcare, "Triage" is the process of determining the priority of patients' treatments based on the severity of their condition. In many public hospitals, the manual triage process is slow, leading to "Red" (Emergency) patients waiting too long. My project aims to solve this by creating a **Simple Reflex Agent** that provides instant, logic-based categorization.

II. System Architecture (PEAS)

To design this agent, I utilized the PEAS framework learned in **Module 1**:

- **Performance Measure:** Minimizing the time taken to identify a critical patient.
- **Environment:** The triage desk where symptom data is entered.
- **Actuators:** The screen output which provides a "Directive" (e.g., "Dispatch Ambulance").
- **Sensors:** The keyboard input used to perceive the patient's state.

III. Logic & Implementation

The agent operates on a **Condition-Action Rule** (If -> Then).

1. The agent first checks for life-threatening symptoms (Breathing/Chest Pain). This ensures that critical cases are never missed.
2. It then uses "Nested Logic" to check for secondary combinations, such as Fever + Fatigue, to identify acute but non-critical cases.
3. Finally, it defaults to a "Stable" status if no symptoms match.

IV. Challenges Faced

- **Input Validation:** Initially, the agent crashed if a user typed "maybe." I solved this by creating a robust input loop that forces a "yes" or "no" response.
- **Logic Hierarchy:** Deciding which symptom was the most important was a challenge. I researched medical triage basics to ensure "Breathing" always took top priority in the code.

V. Conclusion & Learning

This project taught me how to translate a real-world decision-making process into a computer algorithm. While simple, it demonstrates the power of **Symbolic AI** and Knowledge Representation. In the future, this could be expanded using Machine Learning to predict outcomes based on thousands of historical patient records.

Code:

```python
def display_header():
    print("+" + "-"*48 + "+")
    print("|        VIRTUAL MEDICAL TRIAGE ASSISTANT        |")
    print("|          Powered by Logic-Based AI             |")
    print("+" + "-"*48 + "+")
    print("\nINSTRUCTIONS: Please answer 'yes' or 'no' to help the")
    print("agent determine the urgency of the case.\n")

def get_input(prompt):
    while True:
        val = input(prompt).lower().strip()
        if val in ['yes', 'no', 'y', 'n']:
            return 'yes' if val in ['yes', 'y'] else 'no'
        print("Invalid input. Please type 'yes' or 'no'.")

def triage_agent():
    display_header()

    breath = get_input("Q1: Is there severe shortness of breath? ")
    chest  = get_input("Q2: Is the patient feeling heavy chest pain? ")
    fever  = get_input("Q3: Is the body temperature above 102°F (High)? ")
    cough  = get_input("Q4: Is there a persistent dry cough? ")
    fatigue = get_input("Q5: Is the patient experiencing extreme fatigue? ")

    if breath == 'yes' or chest == 'yes':
        # Critical priority - Life-threatening
        status = "CRITICAL (EMERGENCY)"
        action = "Dispatch Ambulance / Move to ICU immediately."
        priority_code = "RED"
    elif fever == 'yes' and fatigue == 'yes':
        # Urgent priority - Needs medical intervention
        status = "URGENT (ACUTE CARE)"
        action = "Schedule General Physician consultation within 4 hours."
        priority_code = "ORANGE"
    elif cough == 'yes' or fever == 'yes':
        # Standard priority - Monitoring required
        status = "NON-URGENT (OBSERVATION)"
```

```python
    elif cough == 'yes' or fever == 'yes':
        # Standard priority - Monitoring required
        status = "NON-URGENT (OBSERVATION)"
        action = "Recommend home isolation and paracetamol. Re-check in 24h."
        priority_code = "YELLOW"
    else:
        # Lowest priority - Minor or no issues
        status = "STABLE (ROUTINE)"
        action = "No immediate threat. Continue standard hydration and rest."
        priority_code = "GREEN"

    print("\n" + "="*50)
    print(f" FINAL DIAGNOSIS: {status}")
    print(f" PRIORITY COLOR : {priority_code}")
    print(f" DIRECTIVE      : {action}")
    print("="*50)
    print("\nDISCLAIMER: This is an AI simulation and not a substitute")
    print("for professional medical advice.")

if __name__ == "__main__":
    triage_agent()
```

Output:

```
+-------------------------------------------------+
|        VIRTUAL MEDICAL TRIAGE ASSISTANT         |
|          Powered by Logic-Based AI              |
+-------------------------------------------------+

INSTRUCTIONS: Please answer 'yes' or 'no' to help the
agent determine the urgency of the case.

Q1: Is there severe shortness of breath? yes
Q2: Is the patient feeling heavy chest pain? no
Q3: Is the body temperature above 102°F (High)? no
Q4: Is there a persistent dry cough? no
Q5: Is the patient experiencing extreme fatigue? yes

==================================================
 FINAL DIAGNOSIS: CRITICAL (EMERGENCY)
 PRIORITY COLOR : RED
 DIRECTIVE      : Dispatch Ambulance / Move to ICU immediately.
==================================================

DISCLAIMER: This is an AI simulation and not a substitute
```